Stage 3 and Stage 4 of Kawasaki Syndrome (KS)

In stage 3 (28 to 31 days), the acute inflammatory process regresses, and granulation tissue forms in the coronary arteries. In stage 4 (40 days to 4 years), scar formation, organization of thrombi, intimal proliferation, and neoangiogenesis continue for several years after disease onset. These changes can lead to severe stenosis of the coronary arteries and other medium-sized arteries.

Although coronary arteries are invariably involved in KS, nearly any medium- or large-sized blood vessel may be affected, as observed at autopsy. Aneurysms have been reported in the femoral, iliac, renal, axillary, and brachial arteries. KS is a systemic vasculitis that primarily affects small- and medium-sized blood vessels. However, cutaneous vasculitis is not a prominent feature.

Diagnosis and Differential Diagnosis

According to the American Heart Association guidelines, the diagnosis of KS requires:

- Fever lasting ≥5 days without another explanation, **and**
- At least four of the following five criteria: 1. Bilateral nonexudative conjunctival injection
  2. Mucous membrane changes (e.g., cracked or injected lips, diffuse oral/pharyngeal erythema, or "strawberry tongue")
  3. Peripheral extremity changes (e.g., palm/sole erythema, hand/foot edema in acute phase, or periungual desquamation in convalescent phase)
  4. Polymorphous exanthem
  5. Acute non-suppurative cervical lymphadenopathy (>1.5 cm in diameter)

In **incomplete KS**, the diagnosis is considered in patients with fever ≥5 days and fewer than four criteria, especially when coronary artery abnormalities are detected by two-dimensional echocardiography or coronary angiography. In some cases with prominent clinical features, treatment may be initiated before day 5 of fever.

IVIG and aspirin (ASA) should be started as soon as the diagnosis is suspected.

Incomplete KS

- Most commonly seen in infants <6 months of age, particularly males
- Defined as fever ≥ 5 days with **at least two** principal clinical criteria
- No other plausible explanation for the illness
- Laboratory findings resemble those of classic KS (e.g., elevated acute phase reactants)
- High risk of coronary artery abnormalities

The diagnosis should be considered in any infant with unexplained fever ≥ 5 days, even with only one clinical criterion.

Findings that suggest an alternative diagnosis include: - Discrete intraoral ulcers - Exudative conjunctivitis - Generalized lymphadenopathy (rare in acute KS)

Differential Diagnosis

KS shares clinical features with infections, hypersensitivity reactions, and environmental toxin exposure. Its mucocutaneous manifestations overlap with other conditions, making differential diagnosis challenging.

A site-specific differential diagnostic approach is useful. The diagnosis relies on recognizing the **cumulative pattern** of mucocutaneous, systemic, and laboratory findings.

Factors that help exclude other conditions: - Measles, influenza: consider local epidemiology, vaccination status, and culture results - Adenovirus infection: can mimic KS, especially in young children

Diagnostic uncertainty may arise in atypical cases that do not meet full criteria but develop coronary artery abnormalities.

Support for initiating treatment in borderline cases may include: - Elevated acute phase reactants (e.g., WBC, ESR) - Pericardial effusion on ultrasound - Anterior uveitis on slit-lamp examination

Treatment

The goal of treatment is to: - Reduce inflammation in the myocardium and coronary artery walls - Prevent coronary thrombosis

Standard therapy for nearly 20 years has been the combination of: - **Intravenous immunoglobulin (IVIG)** - **Aspirin (ASA)**